This assignment will not be graded and is only for practice.

Key Points: 1

Gradient of a function: If f is a scalar valued function defined on a domain D in R^n, then the gradient ∇f is defined as the vector valued function $(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n})$.

If ∇f exists and is continuous in an open ball around the point $(a_1, a_2, \ldots, a_n)$, then the directional derivative of the function f at $(a_1, a_2, \ldots, a_n)$ in the direction of a vector v is given by the dot product of the gradient of the function at $(a_1, a_2, \ldots, a_n)$ with the unit vector in the direction of v i.e.,

$$D_v f(a_1, a_2, \ldots, a_n) = \nabla f(a_1, a_2, \ldots, a_n).\frac{v}{||v||}$$

If ∇f exists and is continuous in an open ball around the point $(a_1, a_2, \ldots, a_n)$, then the gradient is the direction in which the directional derivative at P is maximized amongst all directions, i.e., the directional derivative at P is maximized amongst all the directions in the direction of the vector $\frac{\nabla f(a_1,a_2,\ldots,a_n)}{||\nabla f(a_1,a_2,\ldots,a_n)||}$.

If ∇f exists and is continuous in an open ball around the point $(a_1, a_2, \ldots, a_n)$, then the direction opposite to the gradient is the direction in which the directional derivative at P is minimized amongst all directions, i.e., the directional derivative at P is minimized amongst all the direction in the direction of the vector $\frac{-\nabla f(a_1,a_2,\ldots,a_n)}{||\nabla f(a_1,a_2,\ldots,a_n)||}$.

If ∇f exists and is continuous in an open ball around the point $(a_1, a_2, \ldots, a_n)$, then the maximum value amongst all directional derivatives at P is

$$\nabla f(a_1, a_2, \ldots, a_n).\frac{\nabla f(a_1,a_2,\ldots,a_n)}{||\nabla f(a_1,a_2,\ldots,a_n)||} = \frac{||\nabla f(a_1,a_2,\ldots,a_n)||^2}{||\nabla f(a_1,a_2,\ldots,a_n)||} = ||\nabla f(a_1, a_2, \ldots, a_n)||$$

Similarly, If ∇f exists and is continuous in an open ball around the point $(a_1, a_2, \ldots, a_n)$, then the minimum value amongst all directional derivatives at P is

$$\nabla f(a_1, a_2, \ldots, a_n).\frac{-\nabla f(a_1,a_2,\ldots,a_n)}{||\nabla f(a_1,a_2,\ldots,a_n)||} = -\frac{||\nabla f(a_1,a_2,\ldots,a_n)||^2}{||\nabla f(a_1,a_2,\ldots,a_n)||} = -||\nabla f(a_1, a_2, \ldots, a_n)||$$

1) Find the gradient of the scalar valued function $f(x, y, z) = 2x^3 + xy^2 + 6z^2 + y^2 + xyz$. ***1 point***

- $(6x^2, 2y, 12z)$
- $(6x^2 + y^2 + yz, 2xy + 2y + xz, 12z + xy)$
- $(6x^2 + y^2, 2xy + 2y, 12z)$
- $(6x^2 + yz, 2y + xz, 12z + xy)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(6x^2 + y^2 + yz, 2xy + 2y + xz, 12z + xy)$

2) Find the directional derivative of $f(x, y, z) = 2x^3 + xy^2 + 6z^2 + y^2 + xyz$ at $(2, 1, 2)$ in the direction of a vector $(0, 3, 0)$. ***1 point***

- 30
- 0
- 3
- 10

No, the answer is incorrect.
Score: 0

Feedback:

$$\text{Observe that } \nabla f(2,1,2) = \left(f_x(2,1,2), f_y(2,1,2), f_z(2,1,2)\right)$$

Accepted Answers:

10

3) Find the direction in which the function $f(x, y, z) = 2x^3 + xy^2 + 6z^2 + y^2 + xyz$ increases the fastest from the point $(1, 1, 1)$? ***1 point***

- $(1, 1, 1)$
- $(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$
- $(8, 5, 13)$
- $(\frac{8}{\sqrt{258}}, \frac{5}{\sqrt{258}}, \frac{13}{\sqrt{258}})$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(\frac{8}{\sqrt{258}}, \frac{5}{\sqrt{258}}, \frac{13}{\sqrt{258}})$

4) Find the maximum directional derivative of the function $f(x, y, z) = 2x^3 + xy^2 + 6z^2 + y^2 + xyz$ at the point $P = (1, 2, 3)$. **1 point**

- $\sqrt{1821}$
- 43
- $\sqrt{14}$
- 155

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\sqrt{1821}$

5) Find the minimum directional derivative of the function $f(x, y, z) = 2x^3 + xy^2 + 6z^2 + y^2 + xyz$ at the point $P = (1, 0, 0)$. **1 point**

- -1
- 6
- -6
- 1

No, the answer is incorrect.
Score: 0

Accepted Answers:

-6

6) Let $f(x, y) = x^2y^2 + y^3$. Find the directional derivative of f at the point $(1, 1)$ in the direction which forms an angle (counterclockwise) of $\frac{\pi}{3}$ with the positive x-axis. **1 point**

- $\sqrt{29}$
- $\sqrt{29} + \sqrt{2}$

○ $\sqrt{2}$

○ $1 + \frac{5\sqrt{3}}{2}$

No, the answer is incorrect.

Score: 0

Feedback:

$$\text{The unit vector in the given direction is } \left(\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}\right)$$

Accepted Answers:

$1 + \frac{5\sqrt{3}}{2}$

Key Points: 2

Let $f(x, y)$ be a function defined on a domain D in $\mathbb{R}^2$ containing some open ball around the point (a, b). Equations of the tangent lines to the function f at the point (a, b) in the direction of unit vector (u_1, u_2):

i)**Parametric equations:** $x(t) = a + tu_1, y(t) = b + tu_2, z(t) = f(a, b) + tf_u(a, b)$.
ii) **Symmetric equations:** $\frac{x-a}{u_1} = \frac{y-b}{u_2} = \frac{z-f(a,b)}{f_u(a,b)}$.
iii)**Vector form:** $(x(t), y(t), z(t)) = (a, b, f(a, b)) + t(u_1, u_2, f_u(a, b))$.

The equation of the tangent plane of the graph of a function of two variables $z = f(x, y)$ at the point (x_0, y_0, z_0) (where, $z_0 = f(x_0, y_0)$) is given by

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - 1(z - z_0) = 0$$

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + f(x_0, y_0) = z$$

Linear approximation of the function $z = f(x, y)$ at the point (x_0, y_0, z_0) is given by

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Let $z = f(\underset{\sim}{x}) = f(x_1, x_2, \ldots, x_n)$ be a function defined on a domain D in $\mathbb{R}^2$ containing some open ball around the point $\underset{\sim}{a} = (a_1, a_2, \ldots, a_n)$. Equations of the tangent lines to the function f at the point $\underset{\sim}{a}$ in the direction of unit vector $\underset{\sim}{u} = (u_1, u_2, \ldots, u_n)$:

i)**Parametric equations:** $x_i(t) = a_i + tu_i, z(t) = f(\underset{\sim}{a}) + tf_{\underset{\sim}{u}}(\underset{\sim}{a})$.

ii)**Vector form:** $(\underset{\sim}{x}(t), z(t)) = (\underset{\sim}{a}, f(\underset{\sim}{a})) + t(\underset{\sim}{u}, f_{\underset{\sim}{u}}(\underset{\sim}{a}))$.

Let $f(\underset{\sim}{x})$ be a function defined on domain D in $\mathbb{R}^n$ containing some open ball around the point $\underset{\sim}{a}$. Suppose ∇f exists and is continuous on some open ball around the point $\underset{\sim}{a}$. Equation of tangent plane to f at $\underset{\sim}{a}$:

$$z = f(\underset{\sim}{a}) + \nabla f(\underset{\sim}{a}).(\underset{\sim}{x} - \underset{\sim}{a}).$$

$$z = f(\underset{\sim}{a}) + \sum \frac{\partial f}{\partial x_i}(\underset{\sim}{a})(x_i - a_i)$$

7) Find the equation of the tangent plane to the graph of the function given by $z = f(x, y) = \sqrt{r^2 - x^2 - y^2}$ at the point $(0, 0, r)$. **1 point**

- $x + y + z = r$
- $2x + 2y + 2z = r$
- $z = r$
- $x + y = r$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$z = r$

8) Find the equation of the tangent plane to the graph of the function given by $z = f(x, y) = \frac{1}{\sqrt{c}}\sqrt{d - ax^2 - by^2}$ at the point $(1, 2, f(1, 2))$. **1 point**

- $ax + 2by + \sqrt{c(d - a - 4b)}\ z = 0$
- $ax + 2by + \sqrt{c(d - a - 4b)}\ z = d$
- $2ax + 4by + \sqrt{c(d - a - 4b)}\ z = d$
- $ax + 2by + \sqrt{c(d + a + 4b)}\ z = d$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$ax + 2by + \sqrt{c(d - a - 4b)}\ z = d$

9) Find the equation of the tangent plane to the graph of the function $z = f(x, y) = x^2 y$ at the point $(1, 1, 1)$. **1 point**

- $2x + y - z = 2$
- $2x + y + z = 2$
- $2x + y - z = 0$
- $2x + y = 2$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$2x + y - z = 2$

Key Points: 3

Definition: Consider a scalar valued multivariable function $f : \mathbb{R}^n \to \mathbb{R}$. A point $(a_1, a_2, \ldots, a_n) \in \mathbb{R}^n$ is said to be a critical point of the function f, if $\nabla f(a_1, a_2, \ldots, a_n) = 0$ or at least one of $f_{x_i}(a_1, a_2, \ldots, a_n)$ for $i = 1, 2, \ldots, n$ does not exist.
Steps to find critical points of a function,

i) Find the partial derivatives f_{x_i} for $i = 1, 2, \ldots, n$. If one of $f_{x_i}(a_1, a_2, \ldots, a_n)$ for $i = 1, 2, \ldots, n$ does not exist then $(a_1, a_2, \ldots, a_n)$ is a critical point.

ii) Otherwise, find the all possible point $(a_1, a_2, \ldots, a_n) \in \mathbb{R}^n$ such that $f_{x_i}(a_1, a_2, \ldots, a_n) = 0$ for all $i = 1, 2, \ldots n$, simultaneously.

A function $f(x_1, x_2, \ldots, x_n)$ has a local minimum at the point $(a_1, a_2, \ldots, a_n)$ if $f(x_1, x_2, \ldots, x_n) \geq f(a_1, a_2, \ldots, a_n)$ for all points $(x_1, x_2, \ldots, x_n)$ in some open ball around $(a_1, a_2, \ldots, a_n)$.
A function $f(x_1, x_2, \ldots, x_n)$ has a local maximum at the point $(a_1, a_2, \ldots, a_n)$ if $f(x_1, x_2, \ldots, x_n) \leq f(a_1, a_2, \ldots, a_n)$ for all points $(x_1, x_2, \ldots, x_n)$ in some open ball around $(a_1, a_2, \ldots, a_n)$.
Local extrema are critical points.
Not all critical points are local extrema. Those critical points at which the gradient exists and is 0 but are not local extrema, are called saddle points.
A global (or absolute) maximum/minimum is a point for which the function value is maximum/ minimum amongst all points in the domain. It always exists when the domain is closed and bounded and can be found by comparing the function values at all critical points in the domain as well as on successive boundaries.

10) Let $f(x, y) = \frac{x^3}{3} + y^2 + 2xy - 6x - 3y + 4$. Which of the following options are true about f? **1 point**

- [ ] $f_y = 0$ is a straight line passing through the origin.
- [ ] $(-1, \frac{5}{2})$ and $(3, -\frac{3}{2})$ are critical points.
- [ ] $f_x(-1, \frac{5}{2}) = 0$
- [ ] $f_y(3, -\frac{3}{2}) = 2$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(-1, \frac{5}{2})$ and $(3, -\frac{3}{2})$ are critical points.
$f_x(-1, \frac{5}{2}) = 0$

---

11) Choose the correct statement(s) about the maxima/minima of $f(x, y) = x^4 + y^4$. ***1 point***

- [ ] There are no critical points.
- [ ] There is exactly one critical point.
- [ ] $(0, 0)$ is a point of local minima.

No, the answer is incorrect.
Score: 0

Feedback:

- Observe that $\nabla f(0,0) = 0$ and $f(x,y) \geq 0$ for all $(x,y) \in \mathbb{R}^2$

Accepted Answers:

There is exactly one critical point.
$(0, 0)$ is a point of local minima.

---

12) Suppose $f(x, y) = x^2 + 2y^2 + 3xy - 5y$ be a scalar valued function. Which of the following is (are) critical point(s) of $f(x, y)$? ***1 point***

- ( ) $(-10, 15)$
- ( ) $(0, 0)$

- ○ $(15, -10)$
- ○ $(15, 10)$

No, the answer is incorrect.
Score: 0

Feedback:

$f_x(x,y)=2x+3y$ and $f_y(x,y)=3x+4y-5$

Accepted Answers:

$(15, -10)$

13) Which of the following statements are correct? **1 point**

- ☐ The minimum value of the function $f(x, y) = x^2 + y^2 + 8$ is 0.
- ☐ The minimum value of the function $f(x, y) = x^2 + y^2 + 8 - 4x - 4y$ is 0.
- ☐ The maximum value of the function $f(x, y) = -x^2 - y^2$ is 0.
- ☐ The maximum value of the function $f(x, y) = x^2 + y^2 - 2x + 1$ is 0.

No, the answer is incorrect.
Score: 0

Accepted Answers:

The minimum value of the function $f(x, y) = x^2 + y^2 + 8 - 4x - 4y$ is 0.
The maximum value of the function $f(x, y) = -x^2 - y^2$ is 0.

14) Find the point on the plane $2x - y + 2z = 16$ closest to the origin. **1 point**

- ○ $(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$
- ○ $(1, -1, \frac{13}{2})$
- ○ $(\frac{32}{9}, 1, \frac{89}{18})$
- ○ $(1, 0, -1)$

No, the answer is incorrect.

Score: 0

Feedback:

- Formulate the problem mathematically : find point (x, y, z) on $2x-y+2z=16$ such that distance of (x, y, z) from $(0,0,0)$, i.e., $f(x, y, z)=\sqrt{x^2+y^2+z^2}$ is minimum.
- minimize $f(x, y, z)$ is equivalent to minimize $x^2+y^2+z^2$.
- want to minimize $x^2+y^2+z^2$ with respect to a condition $2x-y+2z=16$.
- Substitute $y=2x+2z-16$ in $x^2+y^2+z^2$ to convert it to a two variable problem.
- Find critical points of $g(x, z)=x^2+(2x+2z-16)^2+z^2$. $(\frac{32}{9}, \frac{32}{9})$ is the only critical point of g.
- So, $(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$ is either a point of local maxima or a point of local minima of $f(x, y, z)$. Value of f at $(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$ is $\frac{16}{3}$.

- If value of f at an arbitrary point on the plane is greater than $\frac{16}{3}$, then $(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$ is a point of minima. For example, $(1, 1, \frac{15}{2})$ is a point on the plane and value of $f(1, 1, \frac{15}{2})=\frac{\sqrt{233}}{2} \geq \frac{16}{3}=f(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$. Thus $(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$ is a point of minima, i.e., the point on $2x-y+2z=16$ which is closest to the origin is $(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$.
- For an arbitrary point, this can be checked by showing that the line joining the origin to $(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$ is orthogonal to the plane and invoking Pythagoras' theorem.

Accepted Answers:

$(\frac{32}{9}, -\frac{16}{9}, \frac{32}{9})$

Your score is: 0/14